Assignment 6: Advanced Statistical Machine Learning

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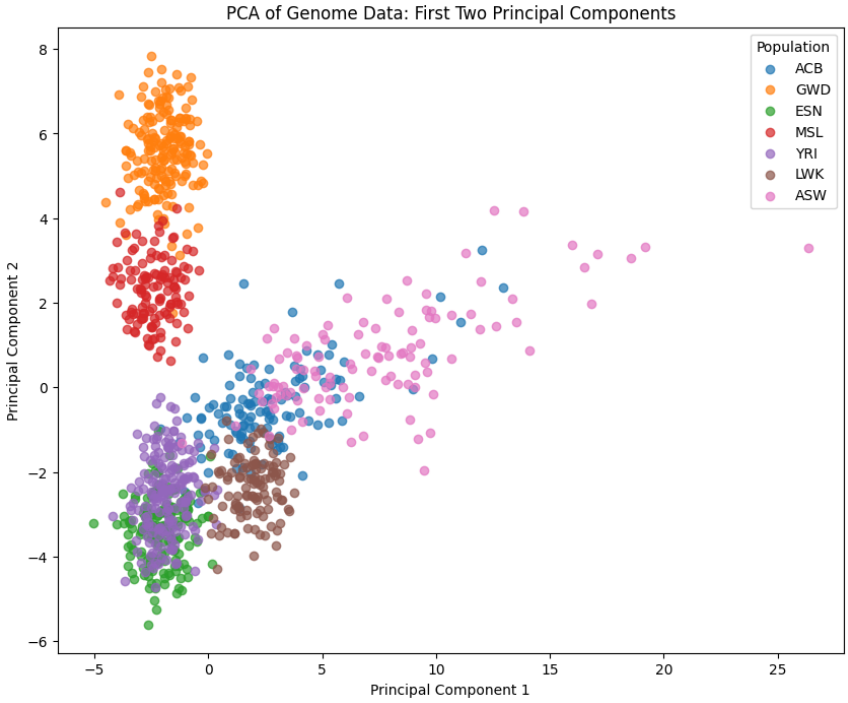
QUESTION 1

Sub question 1

The binary matrix X has 995 individuals. So when PCA is done, the max number of principal components is 995(it can't be more than the number of people).

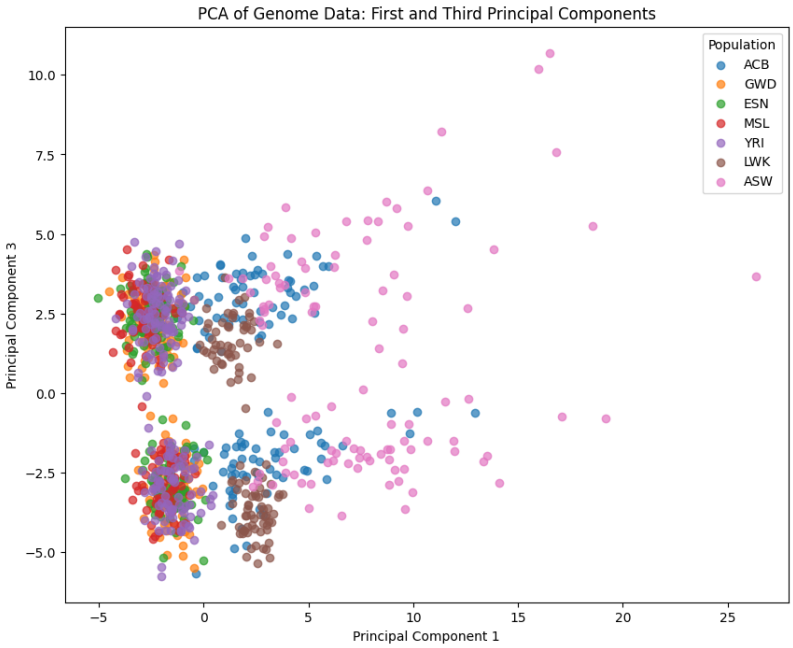
Sub question 2



Sub question 3

The first two principal components show a lot of variation in the dataset. It could be said that population from the same region geographically are more close together. So the PCA shows how the genetics are separated based on where the individuals are from.

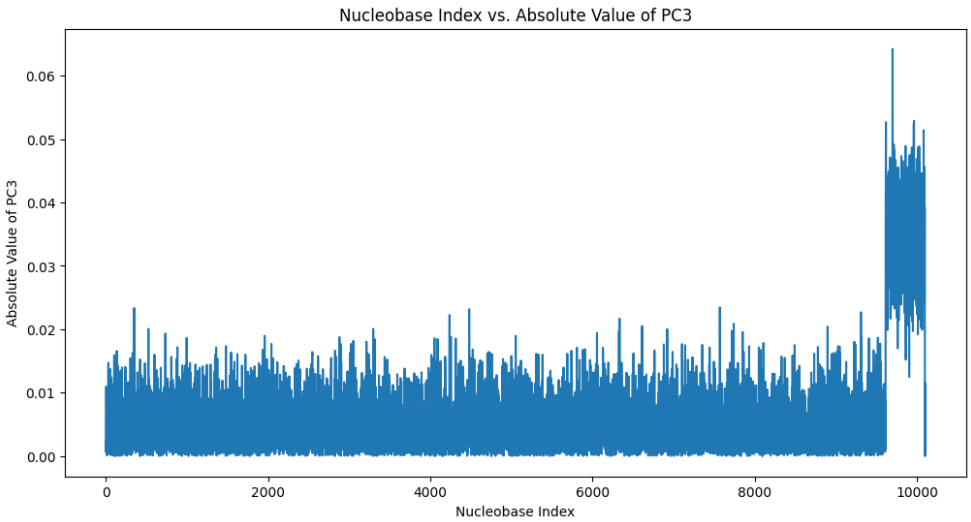
Sub Question 4



Sub Question 5

The third principal component maybe represents some more specific characteristics of genetics such as how it changes over time or if there is any more diversity within a particular geographic region.

Sub question 6

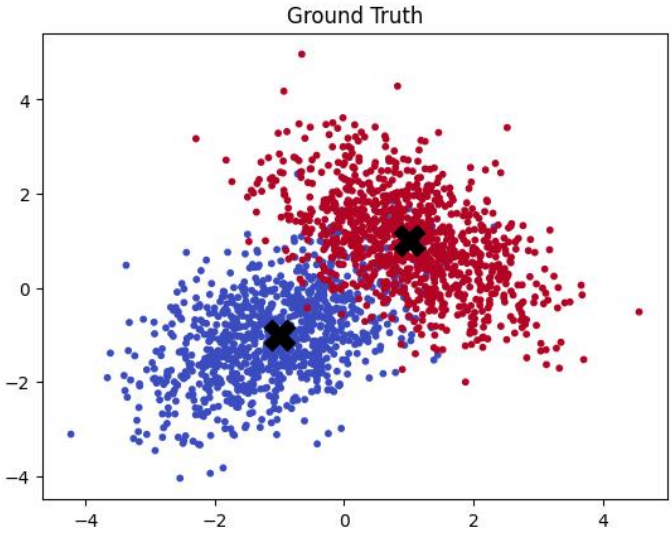


Peaks in the plot show which genomic positions have larger contribution to the 3rd principal component. They're probably due to genetic variations in certain individuals.

Question 2

Sub question 1

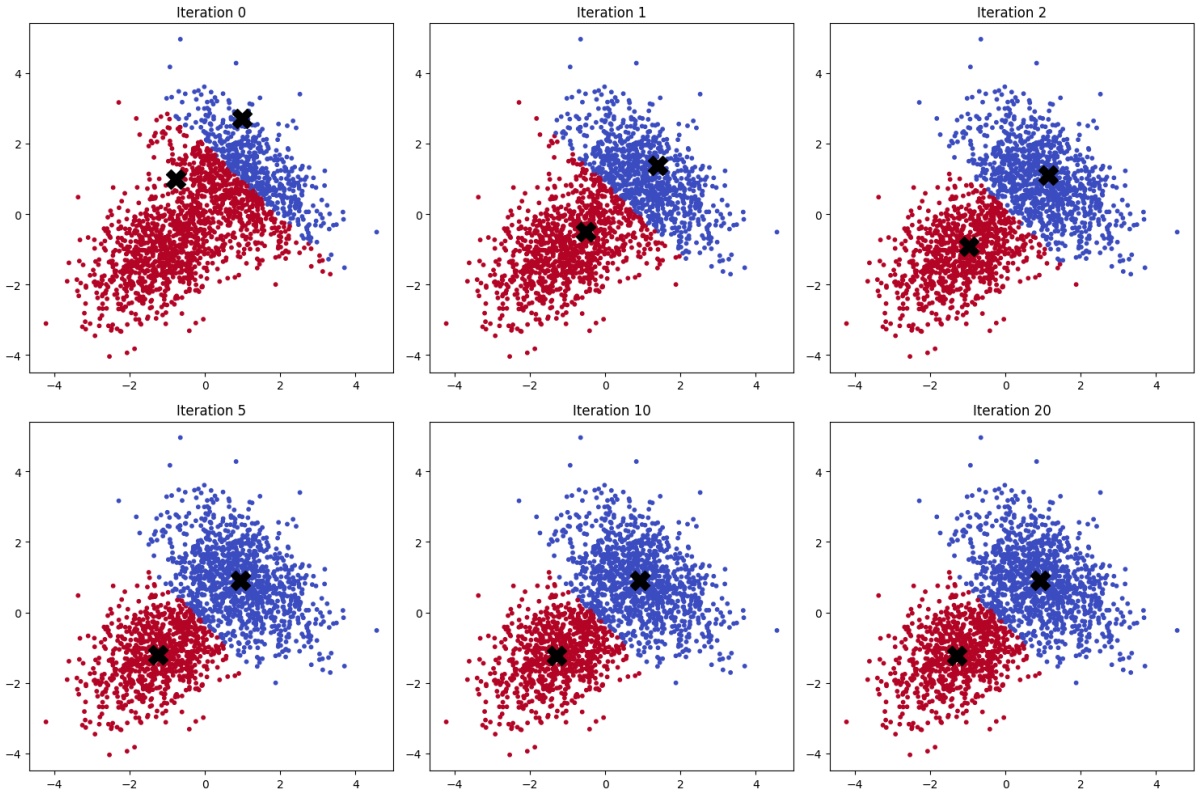
Ground Truth cluster labels and means are plotted correctly.



Sub question 2

k-means ARI: 0.704

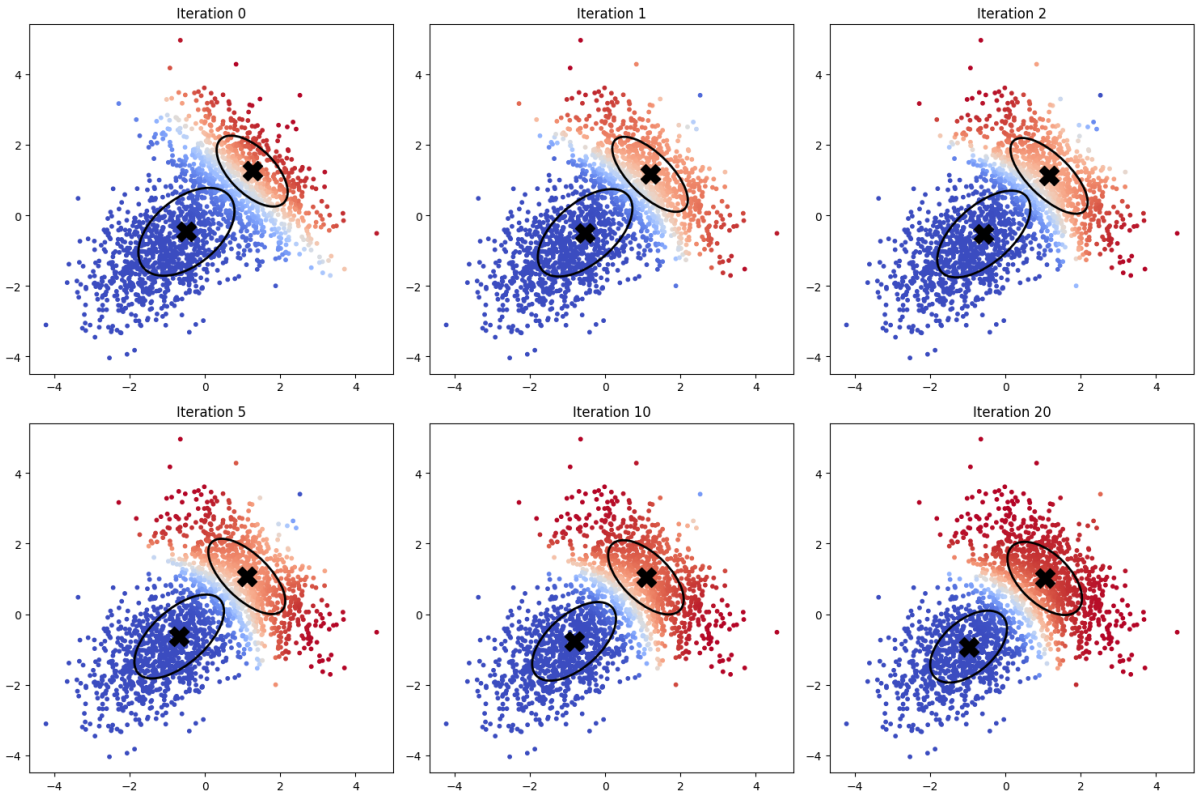
k-means for different iterations:



Sub Question 3:

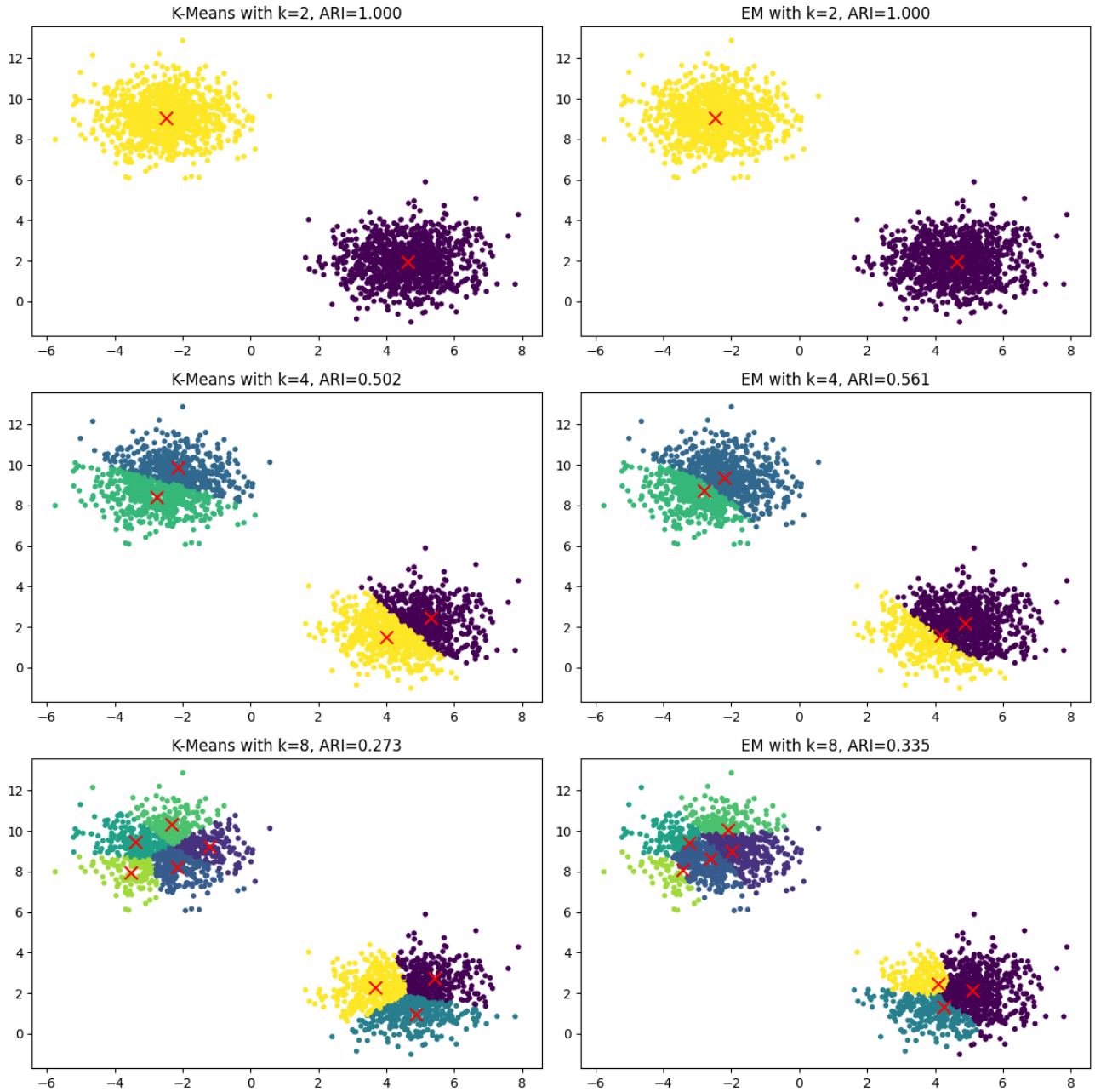
EM ARI: 0.769

EM for different iterations:



Sub question 4:

Comparison between K-means and EM -



1.

Usually, the EM performs better than K-means when clusters are not spherical or overlap, because EM uses Gaussian assumptions and probabilistic assignments. But sometimes, for spherical clusters, both have a similar performance.

From the above code, we get K-Means ARI: 0.704 and EM ARI: 0.769.

This indicates that EM performed slightly better in handling overlapping clusters.

2.

K-Means:

a. Strengths: It is easy to impliment and fast. It performs better for spherical and well-seperated clusters. It is also able to handle large datasets.

b. Limitations: It can't perform well with overlapping clusters. It is sensitive to outliers.

EM:

a. Strengths: It is able to handle overlapping clusters better than K-means. It is is general a little more flexible than K-means.

b. Limitations: It is computationally more expensive compared to K-means. It always assumes that the clusters have Gaussian distributions which isn't always the case.

3.

K-Means:

It assigns a hard label to each point i.e it fully classifies a point belonging to one cluster. This also can be seen as hard boundaries are there, since convergence abruptly shifts the centroids and points shift to the nearest centroid.

EM:

EM assigns soft labels based on probabilities. This implies that convergence is smooth and slower than k-means. So the elipses adjust iteratively.

Sub question 5

1.

When k>2 -

K-means: The existing clusters are split into smaller sub-clusters. The ARI decreases because the predicted clusters don't align with the ground truth.

EM: It tries to fit the gaussian distributions to smaller regions of the data which produces overlapping clusters.

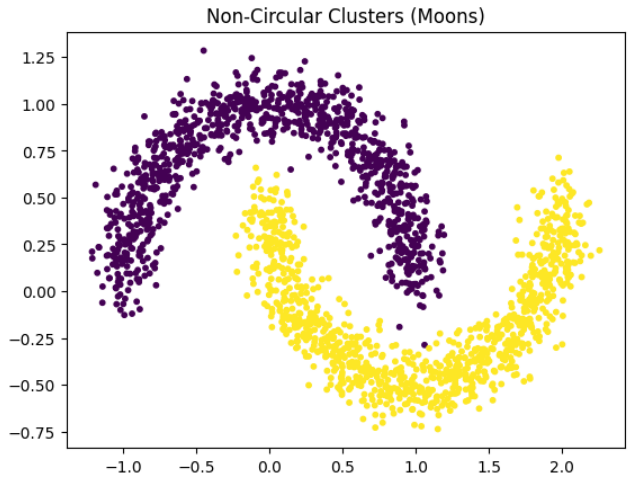
When k= 4,8 -

K-Means: Splits the clusters into smaller sub-clusters.

EM: Models sub-clusters with overlapping Gaussian components.

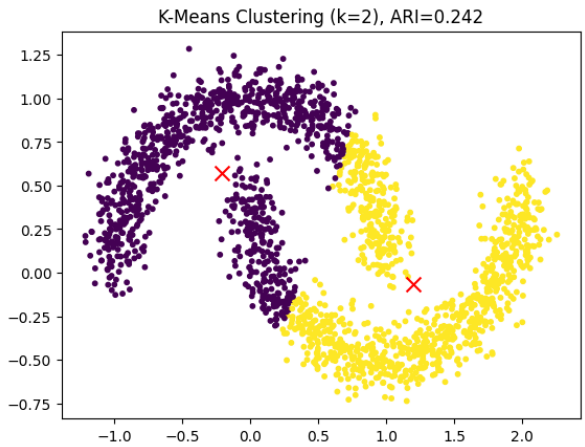
Sub question 6

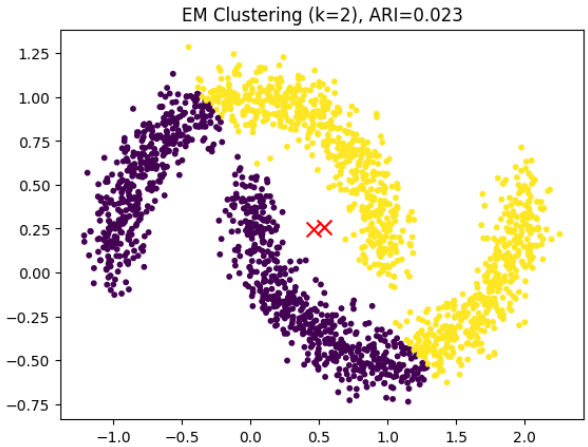
1.



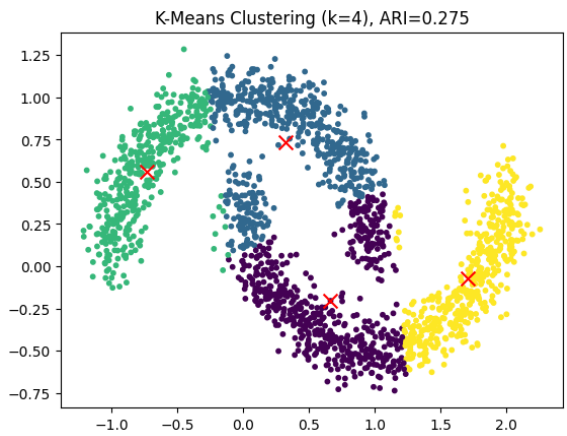
2.

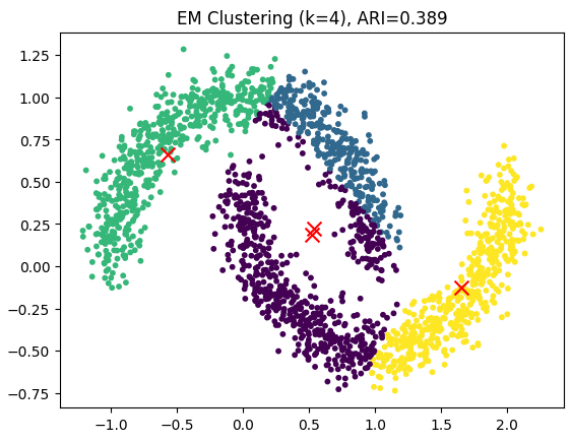
a. K-Means ARI for k=2: 0.242
EM ARI for k=2: 0.023





b. K-Means ARI for k=4: 0.275
EM ARI for k=4: 0.389





3.

K-Means: Performs poorly on non-circular clusters because it assumes spherical shapes. The moon shapes can easily be split with boundaries.

EM: It performs a little better because it deals with probabilities. But EM will also perform badly if the data doesn't use Gaussian distribution.

Conclusion: Both algorithms don't handle non-circular clusters fully. But EM is a tiny bit better than K-means due to its use of soft labels.

4.

To fix this, we can use:

a. DBSCAN (Density-Based Spatial Clustering): It detects clusters based on the density of points. It can handle arbitrarily-shaped clusters easily.

b. Spectral Clustering: It uses graph-based clustering to capture non-linearity in structures in the given data.

c. Gaussian Mixture Models with Kernel Functions: It uses kernel-based Gaussian Mixture Models to capture complex non-linear cluster shapes.

Question 3

1.

2. Same as above
3.
    a. As K increases – the image has more details and slowlt starts looking like the original image. We can also see more colours.
    b. If small K is used: The image looks very simple with nor many colours and details.
       If large K is used: The image is more detailed and we can see many colours. However this also means that it will take longer to compute and more storage space might be needed to save it.